

Process Mining and Formal Verification of the Volvo IT Incident Process

BPI Challenge 2013

Formal Methods

September 8, 2026

1. Context
2. The data
3. Discovery
4. Time
5. Verification
6. Findings
7. Implementation

Three findings are argued for:

- rework dominates the process;
- the documented procedure is not the executed one;
- one classical algorithm is inapplicable here, for a reason theory predicts.

An information system records every state change it performs. That record is an *event log*: for each case, a finite sequence of (*case, activity, timestamp*) events.

Discovery derive a model — here a Petri net — from the log alone.

Conformance measure how well a model and a log agree.

Verification decide whether a stated property holds of every case.

A designed model says how work *should* proceed. A discovered model says how it *did*. The gap between them is the object of study.

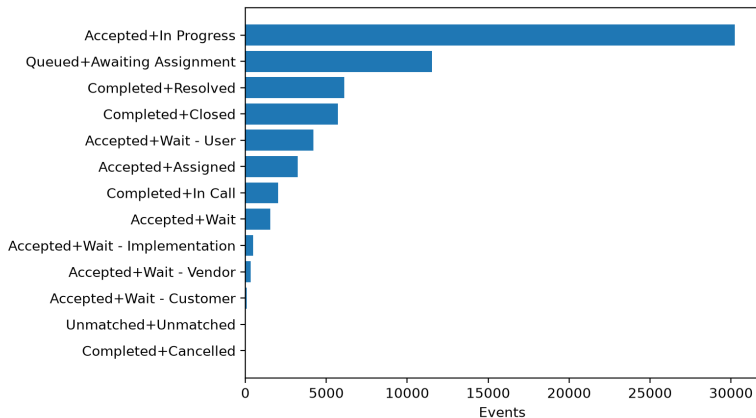
- An IT service desk records every state change of every incident.
- The process is *documented*. What was actually *executed*?
- Volvo IT Belgium, VINST system, 2010–2012.
- Three logs: incidents, open problems, closed problems.
- A standard benchmark, so the results are comparable with the literature.
- Nothing here is read off a dashboard by eye: every figure shown is produced by code and pinned by a regression test.

Log	Cases	Events	Activities	From	To
closed_problems	1,487	6,660	7	2006-01-11	2012-05-31
incidents	7,554	65,533	13	2010-03-31	2012-05-22
open_problems	819	2,351	5	2006-11-07	2012-06-15

Case and event counts reproduce the published figures exactly — the first check on the loader.

The problem logs are not variants of the incident process: median duration differs by an order of magnitude and their alphabet is smaller. The incident log is the object of study; the other two are used for comparison.

One activity carries the process



Accepted+In Progress is the plurality of all events — the first hint that this process spends its time repeating itself.

There is no activity column

The dataset records Status and Sub Status. What counts as an activity is the analyst's decision.

Status — 4 activities

Accepted, Queued, Completed, Unmatched

Small models. Waiting on a vendor and waiting on a user become indistinguishable.

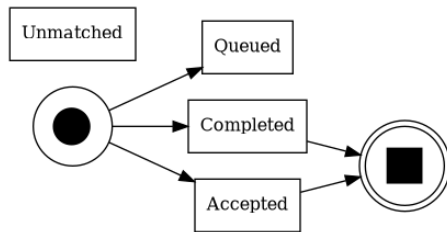
Status+Sub Status — 13 activities

Accepted+In Progress, Accepted+Wait - User, ...

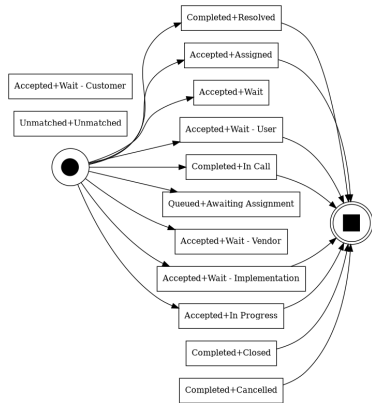
Three times larger models, lower precision, and the real process.

Treated as a parameter of the loader, not fixed silently. Both are reported.

The same miner, the same log, two models



Status — 4 labels



Status+Sub Status — 13 labels

The labelling decision is not cosmetic. It is the decision.

Three discovery algorithms

Alpha Ordering relations from directly-follows pairs. Assumes a complete, noise-free log. Cannot represent short loops.

Heuristics Frequency-weighted, with a dependency threshold. Filters infrequent behaviour instead of modelling it. Robust to noise.

Inductive Recursive partitioning into a process tree, converted to a net. Guarantees soundness and, at zero noise, perfect fitness.

All three are run on the full incident log, under both labellings, from the same loaded bundle — so the six rows on the next slides differ only in the algorithm and the labelling.

Fitness Can the model replay what happened?

Precision Does it forbid what never happened?

Generalisation Will it hold for future cases?

Simplicity Can a human read it?

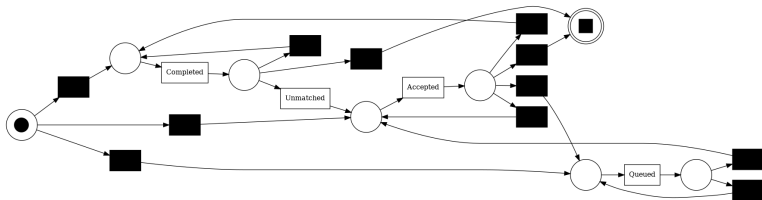
They trade off. Maximising any one alone is trivial — a model permitting everything fits perfectly and explains nothing; a model enumerating the observed traces is perfectly precise and predicts nothing. Hence a table, not a winner.

Mode	Algorithm	Fitness	Precision	Gen.	Simpl.	Places	Trans.
status	alpha	0.119	0.400	0.734	1.000	2	4
status	heuristics	0.994	0.678	0.767	0.622	8	15
status	inductive	1.000	0.440	0.748	0.667	17	23
status+substatus	alpha	0.127	0.176	0.770	0.789	2	13
status+substatus	heuristics	0.948	0.387	0.634	0.471	27	69
status+substatus	inductive	1.000	0.234	0.783	0.634	57	78

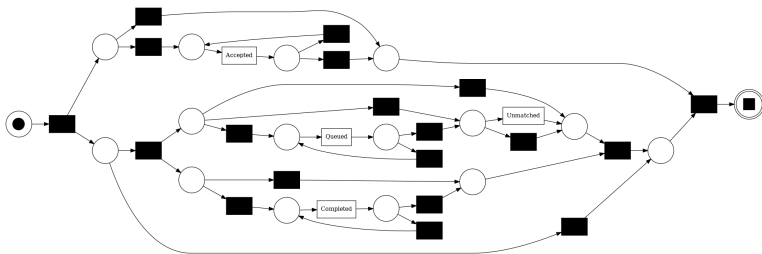
Inductive buys perfect fitness with permissiveness and the largest net. Heuristics gives the best balance. Alpha collapses.

Moving from 4 labels to 13 lowers precision for every algorithm and roughly triples the net — there is simply more distinct behaviour to explain.

What the two usable miners produce

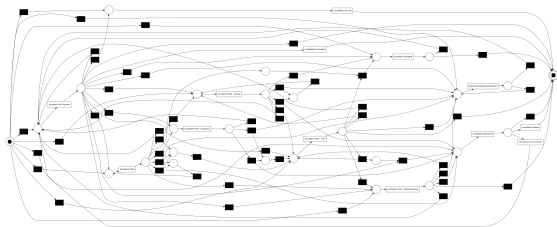


Heuristics — 8 places, 15 transitions

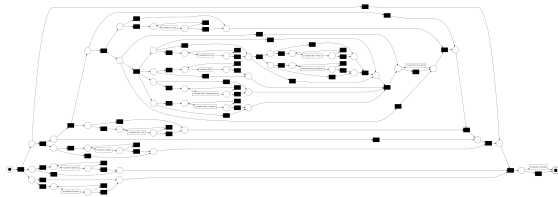


Inductive — 17 places, 23 transitions, plus silent routing

Thirteen labels, and the net stops being readable



Heuristics — 27 places, 69 transitions



Inductive — 57 places, 78 transitions

Simplicity is a conformance dimension for a reason.

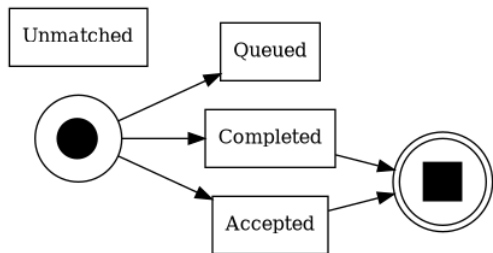
Why Alpha collapses

Fitness below 0.15; the net degenerates to two places. **This is not a bug.** 90% of cases contain a

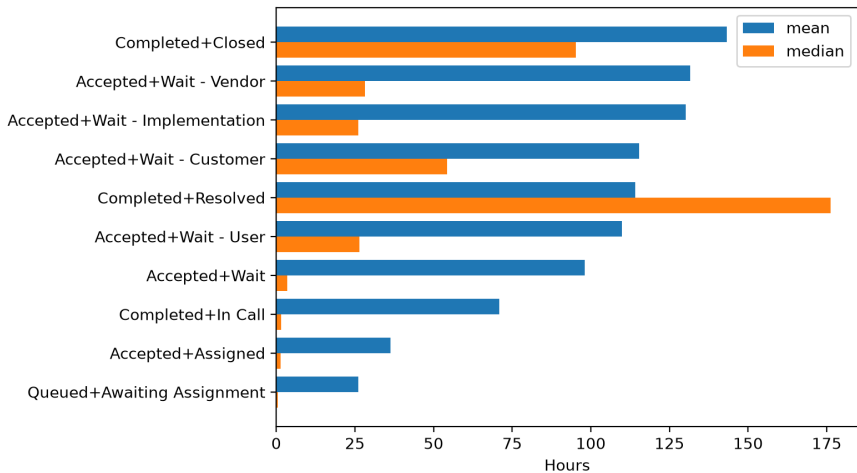
length-one loop — an activity immediately repeated.
Alpha infers contradictory ordering relations from a

directly-follows pair (a, a) and provably cannot represent such loops. A known theoretical boundary,

demonstrated on a real log.



Where does the time go?



The Wait states dominate. Delay is *external* to the service desk.

The same result in numbers

Activity	n	Mean (h)	Median (h)	Max (h)
Completed+Closed	143	143.31	95.28	736.50
Accepted+Wait - Vendor	313	131.63	28.15	1633.67
Accepted+Wait - Implementation	491	130.22	26.00	3500.55
Accepted+Wait - Customer	101	115.45	54.39	1488.93
Completed+Resolved	6,025	114.15	176.25	192.10
Accepted+Wait - User	4,214	109.97	26.36	5059.88
Accepted+Wait	1,533	97.99	3.45	4992.31
Completed+In Call	153	70.88	1.56	677.31
Accepted+Assigned	3,220	36.31	1.36	17334.07
Queued+Awaiting Assignment	11,544	26.16	0.59	6838.52
Accepted+In Progress	30,237	10.63	0.03	7896.28
Unmatched+Unmatched	5	0.19	0.07	0.61

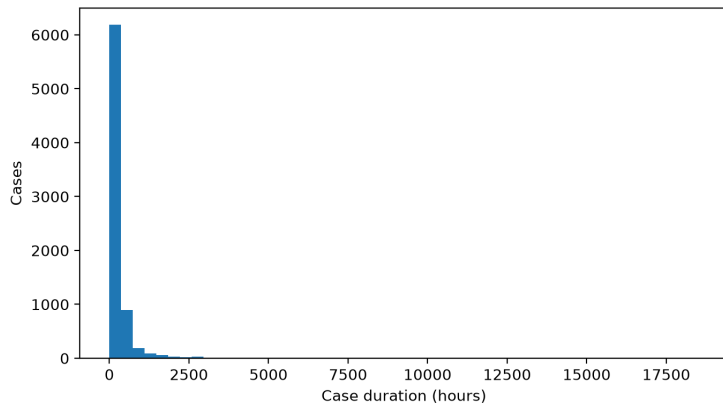
Waiting time is the interval to the next event of the same case; last events are excluded, not counted as zero.

Accepted+In Progress: median waiting time in **minutes**, mean in **hours**, maximum in **months**.

- Work is normally picked up immediately.
- Occasionally it stalls for a very long time.
- The mean alone says “slow activity”. The median alone hides the stalls.

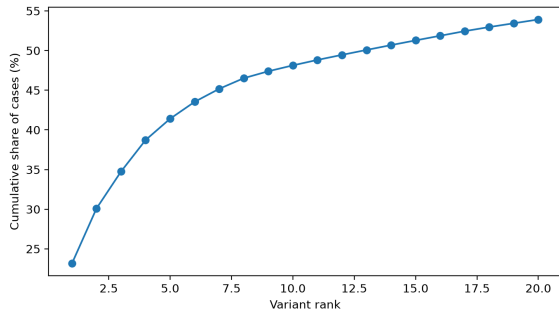
Both are reported everywhere in this analysis, and so is the maximum.

How long a case lasts



Heavy-tailed at the case level too: most incidents close quickly, a minority runs for months. The same shape as the waiting times, one level up.

A weakly standardised process



Thousands of distinct variants against 7,554 cases. Most variants occur exactly

once. Consequence: any high-precision

model of this log would be overfitted. The low precision figures are the correct result.

A case is a finite sequence of events, so the semantics are LTL_f .

$$\pi, i \models F\varphi \iff \exists j \geq i. \pi, j \models \varphi$$

$$\pi, i \models G\varphi \iff \forall j \geq i. \pi, j \models \varphi$$

$$\pi, i \models X\varphi \iff i + 1 < n \textbf{ and } \pi, i+1 \models \varphi$$

X is *strong*: false at the last position. Remember this for two slides' time.

Eight properties, each a real formula evaluated per case.

Termination	$F(\text{completed}_*)$
Resolution	$G(\text{in_progress} \rightarrow F \text{ completed}_*)$
No work after closure	$G(\text{closed} \rightarrow \neg F \text{ in_progress})$
Queued work picked up	$G(\text{queued} \rightarrow F \text{ in_progress})$
Formal closure	$G(\text{in_progress} \rightarrow F \text{ closed})$
Queueing precedes work	$(\neg \text{in_progress}) \cup \text{queued}$

The last two encode the *documented* procedure. They are expected to fail. The size of the gap is the result.

Property	Satisfied	Violated	Ratio
Termination	7,546	8	0.999
Resolution	7,545	9	0.999
No work after closure	7,435	119	0.984
Queued work is picked up	7,494	60	0.992
Waiting on the user resumes	7,550	4	0.999
No unmatched events	7,549	5	0.999
Formal closure	5,591	1,963	0.740
Queueing precedes work	1,168	6,386	0.155

Every case of the incident log, every property. No sampling.

- **Cases do not disappear.** Termination, resolution and recovery from Wait - User each hold for essentially every case.
- **Closure is not universal.** A quarter of worked incidents never reach Completed+Closed; they settle as Completed+In Call. Defect or undocumented route — the analysis establishes only that both exist at scale.
- **Queueing before work all but never happens.** The one property stating the documented procedure is the one that fails.

Fewer than half of all events are the *first* occurrence of their activity within the case.

- Over half of all recorded work is repetition.
- Completion is near perfect; variability is moderate.
- Rework is the single component dragging the score down — and the largest available lever.

One score, and what it is made of

Log	Score	Grade	Completion	Rework	Standardisation	Throughput
closed_problems	65.9	D	1.000	0.655	0.780	0.000
incidents	74.4	C	0.999	0.481	0.698	0.748
open_problems	70.0	D	0.435	0.746	0.778	0.942

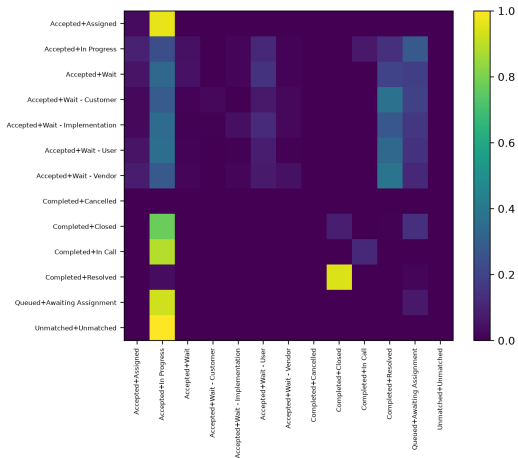
Four normalised components: completion, first-occurrence share, one minus the variant ratio, and throughput against a fixed horizon.

The weights and the horizon are *stipulated*. The components are meaningful; the aggregate only comparatively — and comparison is what it is used for.

$(\neg in_progress) \cup queued$ holds for a small minority of cases.

- The documented procedure: queue the incident, then work it.
- What happens: work begins without the incident ever being queued.
- Established by *formal verification over every case* — not by sampling, not by eyeballing variants.

The same pattern in the transitions



Directly-follows frequencies, normalised per row: given this activity, what comes next?

One column absorbs nearly every row — Accepted+In Progress. Work is what follows everything.

The Queued+Awaiting Assignment *column* stays faint: almost nothing flows *into* the queue. Push to front, visible without a formula.

Verification

$G(\text{closed} \rightarrow \neg F \text{ in_progress})$ fails for a set of cases: work resumes after formal closure.

Performance

Completed+Closed appears in the waiting-time table at all — meaning closure events that were *not* the last event of their case.

Independent methods agreeing is stronger than either alone.

The strong-X trap

Written the natural infinite-trace way:

$$G(\textit{closed} \rightarrow X G \neg \textit{in_progress}) \implies 26.2\%$$

Written without the next operator:

$$G(\textit{closed} \rightarrow \neg F \textit{in_progress}) \implies 98.4\%$$

Most cases *end* with `Completed+Closed`. `X` is false there, so every such case falsifies the first formula for reasons having nothing to do with Volvo IT.

A factor of four apart. Only one of them is about the process.

The packages

`logs` loading, schema detection, filters

`mining` discovery, conformance,
performance

`verification` LTL_f per trace

`analysis` score, anomalies, prediction

`ui`, reporting and scripts sit on top;
nothing depends on `ui`.

Consequences

- Every function takes a loaded log and returns a value — no module state, no cache
- Dashboard callbacks are plain functions, called directly by 202 tests
- Two browser sessions are independent
- Every table and figure on these slides is generated from code

The loader adds columns; it never drops any.

Normalisation *adds* the XES-standard names to a copy of the raw frame. The support-line, ownership and country attributes survive — reducing each event to three columns, as is common, would make them unrecoverable without reparsing.

A named property is not a special case.

The eight properties are declared as data — name, formula, measured ratio — and evaluated by the same function that evaluates a formula typed into the dashboard. The recorded ratios are asserted by the test suite, so a change that moved a published figure would fail the build rather than quietly update the report.

- Conformance is computed on a *sample*; alignment-based computation does not scale to the full log.
- Token-based replay is optimistic on unsound nets. Alignments agree on the sample.
- The composite score's weights are *stipulated*. Components are meaningful; the aggregate only comparatively.
- Unmatched+Unmatched is an export artefact, not work. Retained so its prevalence is visible.
- We observe what was *recorded*. Poor recording discipline looks like rework.

- ① **Rework dominates** — over half of recorded work is repetition.
- ② **Documented $\neq$ executed** — push-to-front proved formally over every case.
- ③ **Alpha is inapplicable here** — predicted by length-one loops, not an empirical surprise.

Next: the organisational attributes are already loaded — handover analysis across support lines, push-to-front quantified by organisation, and incident versus problem as interacting processes.

Questions